Mode of Examination: Online
M.Sc. Semester – I Examination, 2021

2021
Subject: Computer Science
Paper Code & Name: CSMC-102 (Data Structure and Algorithms)
Full Marks: 30
Date: 01.02.2022 **Time and Duration: 12.30 PM – 2:00 PM; 90 Minutes**

Please follow the following instructions carefully:
Promise not to commit any academic dishonesty.
Candidates are required to answer in their own words as far as applicable.
Each page of the answer scripts should have your Full Name (in CAPITALS) on the right-hand side top corner. The script may be a softcopy also.
The name of the scanned copy of the answer script will be of the following format:
CSMC-102-DSA-FirstName-Surname.pdf
(Example: CSMC-102-DSA-Xxx-Yyy.pdf)
The subject of the mail should be the file name only.
The scanned answer script is to be sent to cucse2020@gmail.com
Extra credit for the format of document and presentation.
The report should have an index page.
Extra 10 minutes is allowed to upload the answer script.
The answer script may not be accepted after the scheduled time.

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1. Answer any five from the following: [2×5 = 10]

(a) What is DFS? Discuss its complexity issues.
(b) Clearly state the steps to construct an equivalent binary tree of the following tree: (Z(Y(V,U), X(T), W(S(P), R(O,N), Q))).
(c) Prove that the binary search takes O(log n) time for an ordered list of n elements.
(d) Suppose that you are given three different weights X, Y, and Z. Draw a decision tree to compare the weights in descending order.
(e) Show that the external path length of a 2-tree is minimum when all the leaves of the tree are on the same level or on two adjacent levels.
(f) Show the divide and conquer steps of multiplying the binary numbers 1101 with 0101.
(g) Solve the following recurrence: $f(n) = 9f(n/3) + n^2$, for $n \geq 2$; $f(1) = 1$.

2. Answer any four from the following: [5×4 = 20]

(a) Let A[1 .. 60] be an array of integers. Suppose we run *Select* algorithm (which does not resort to sorting a subarray if its size goes below a certain threshold and always searches the element recursively) to find the 17th smallest element in A. How many calls to *Select* algorithm will there be if the array is sorted? Justify your answer by showing the steps.
(b) You are given an array A[1 .. n] of integers and an integer x. Derive a divide and conquer algorithm to find the frequency of x in A, i.e., the number of times x appears in A. Derive the time complexity of your algorithm.
(c) Obtain the addressing formula for the element $A[i_1, i_2, \ldots, i_n]$ in an array declared as $A[l_1 .. u_1, l_2 .. u_2, \ldots, l_n .. u_n]$. Assume a row-major representation of the array with one word per element and α is the address of $A[l_1, l_2, \ldots, l_n]$. Show that the addressing formula is linear.
(d) Clearly state, if there are any assumptions, and describe how BFS may work to find out one of the shortest paths between a pair of vertices of a graph, with the help of a suitable data structure.
(e) Consider the following sequence and insert the keys, in the order given, to build them into an AVL tree. Show the tree after each insertion. Then, delete j from the final AVL tree obtained.

$$d, s, v, g, y, j, h, m, i, q$$

(f) Draw the following 2-3 tree that comprises a linearly ordered set, S = {{1.1, 2.2, 3.3}, {5.5, 6.6, 7.7}, {8.8, 9.9}}. Obtain the 2-3 tree, S', after inserting 4.4 into S. Also, obtain the 2-3 tree after deleting 8.8 from S'.